DANIEL KARIMI

SOFTWARE DEVELOPMENT ENGINEER INTERN

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Madrid, Spain
daniel.karimi.alvarez@gmail.com
+34 654 310 492
linkedin.com/in/daniel-karimi-alvarez
github.com/Dankar44

ABOUT ME

Final-year Telecommunications Systems Engineering student at **Universidad Politécnica de Madrid** (expected graduation 2025/2026), with a strong foundation in software development, **Object-Oriented Design (Java, Python, C++)**, data structures, and algorithms. I have a proven ability to learn, adapt, and thrive in fast-paced, technically demanding environments.

During my internship at CSIC, I designed, built, and owned the full lifecycle of scalable Python-based software pipelines, working closely with cross-disciplinary teams and applying **CI/CD principles, version control (Git), and operational best practices**. I have hands-on experience integrating **Generative AI and Deep Learning tools** (PyTorch, YOLOv8) to drastically boost development productivity and solve real customer problems — reducing processing times from hours to minutes.

I am eager to grow at Amazon as an SDE Intern, bringing my curiosity, ownership mindset, and ability to write clean, maintainable code to build and operate scalable services that directly impact millions of customers worldwide.

TECHNICAL STACK

SOFTWARE DEVELOPMENT

Java • Python • C / C++ • Object-Oriented Design • Data Structures • Algorithms

CLOUD & DISTRIBUTED SYSTEMS

AWS (Familiarity) • Microservices Concepts • RESTful APIs • Cloud-Native Architecture • Docker • Distributed Systems

GENAI & ML

Generative AI Tools • PyTorch / TensorFlow • ML Pipeline Design • Deep Learning • Optimization • Automation

DEV PRACTICES

Git & GitHub • CI/CD Workflows • Agile / Scrum • Code Reviews • Unit Testing & QA • Technical Documentation

SYSTEMS & OS

Linux / Unix OS • System Architecture • Networking (TCP/IP) • Bash Scripting • Debugging & Troubleshooting

ENGINEERING FOUNDATIONS

Problem Solving • Analytical Thinking • Electronics & Embedded • Microprocessors • Signal Processing

Languages: Spanish (Native) • English (B2 – Aptis ESOL, British Council – excellent written and verbal communication)

PROFESSIONAL EXPERIENCE

Software & AI Engineer Intern | CSIC – Institute of Physical and Information Technologies (ITEFI)

Sep 2025 – Feb 2026

- Took **full ownership** of the design, development, and deployment of scalable Python-based software pipelines for large-scale data processing, applying professional coding standards and design patterns.
- Leveraged **Generative AI and Deep Learning tools** (YOLOv8, PyTorch) to enhance development productivity, automating previously manual analytical workflows and slashing execution times from hours to minutes.
- Participated actively in code planning, written technical documentation, and peer reviews, maintaining clean, maintainable, and reproducible codebases in a Linux/Unix environment.
- Applied **CI/CD principles and Git version control** to manage source code, track changes, and ensure operational excellence across all project stages within a fast-paced research environment.
- Collaborated effectively with cross-functional scientists and engineers, translating complex technical findings into clear written and verbal communication for both technical and non-technical stakeholders.

Python | GenAI Tools | CI/CD | Git | OOP | Linux | Code Reviews

EDUCATION

Telecommunications Systems Engineering (BSc) | **Universidad Politécnica de Madrid (ETSIT-UPM)** *Sep 2020 – Jul 2025/2026 (expected)*

- **Programming & Software Engineering:** Courses in Data Structures and Algorithms, Object-Oriented Programming (OOP) in **Java and C++**, software design patterns, and engineering problem-solving methodologies.
- **Systems & Networking:** TCP/IP architecture, distributed operating systems, Unix/Linux environments, and scalable system design. Solid foundation in debugging and fault-tolerant architectures.
- **Electronics & Embedded Systems:** Microprocessors, digital circuits, hardware-software integration, and low-level programming — highly relevant to embedded and scalable service development.

Java | C/C++ | Python | Algorithms | Linux | Networking | Embedded Systems

Technology Baccalaureate, **IES Isabel la Católica** *2018 – 2020*

Science and Technology track. Honours. Strong mathematical and analytical foundation.

CERTIFICATIONS

AI and Productivity (Google): Effective use of GenAI tools and automation workflows to maximize development productivity. •

Generative AI (MIT): Building and deploying GenAI solutions; leveraging agentic AI technology for scalable automation. •

Internet of Things (MIT): Distributed architectures, sensory data integration, and cloud platform connectivity for scalable IoT systems. •

Aptis ESOL B2 (British Council): Official English certification demonstrating strong professional communication skills for multi-cultural, cross-functional engineering teams.

KEY PROJECTS

CellTracker (End-to-End Software System) OWNED FULL LIFECYCLE

PYTHON • OOP • GENAI • ALGORITHMS

Scalable algorithmic tracking framework — pending publication in scientific journal.

Owned the full development lifecycle of an end-to-end software system integrating GenAI-based detection models with classical algorithmic tracking (Kalman Filters). Designed using object-oriented principles for maximum modularity, scalability, and fault tolerance, deployable across varied research environments.

- Applied agile iteration cycles: rapid prototyping, testing, and continuous integration of feedback from cross-disciplinary team members.
- Maintained comprehensive technical documentation for each stage of the project lifecycle.

MigraCell Accelerator SCALABLE SERVICE

PYTHON • DEEP LEARNING • MICROSERVICE ARCHITECTURE

Automated service leveraging deep learning to eliminate manual bottlenecks in data analysis.

Designed and built a scalable analysis service utilizing Deep Learning (YOLOv8-seg) to automate complex data extraction processes. Architected a clean, modular codebase ensuring fault tolerance and long-term maintainability, reducing turnaround from hours to minutes.

- Leveraged GenAI tools to enhance development speed and applied best practices for writing clean, maintainable code.
- Validated correctness through automated testing and iterative precision metrics evaluation.